

MULTI-COMMODITY MINIMUM-COST FLOW FOR BLOOD DISTRIBUTION IN MALOLOS: A DIRECTED CAPACITY-CONSTRAINED NETWORK MODEL

An Undergraduate Exposition Paper Presented to

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In Partial Fulfillment of the Requirements for
the Course Graph Theory and Its Application under
the Degree Bachelor of Science in Mathematics
with Specialization in Computer Science

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2nd Semester, SY 2025–2026

TABLE OF CONTENTS

Abstract	iv
The Problem and Its Background	1
Background of the Study	1
Statement of the Problem	2
Scope and Limitation	3
Significance of the Study	4
Graph-Theoretic Foundations of the Model	5
Graph Theory Framework	5
Multi-Commodity Flow Model	6
Methodology	8
Description of the Real-Life System	8
Construction of the Graph Model	8
Dataset	8
Edges	10
Modelling Assumptions and Constraints	10
Commodities(blood types)	11
Supply and demand	11
Capacities	11
Cost	11
Decision Variables	11
Flow Conservation Constraints	12
Edge Capacity Constraints	12
Objective Function (Min-Cost with Priority)	12
Applied Graph-Theoretic Algorithms	12
Efficiency in Multi-Objective Optimization	13

Structural Advantages	13
Results and Discussion	14
Example Calculation	14
Example Calculation 1	15
Discussion	20
Summary and Conclusion	21
Summary of Findings	21
Conclusions	21
references	22
Appendix A: Vertex and Edge List	23
Appendix B: SageMath/Python Codes	24
1. The “Multi-Commodity” Approach	25
2. Managing the Supply and Demand Network	25
3. Mathematical Rules (Constraints)	26
4. The Goal: Speed and Priority	26
5. Handling Real-World Disruptions	26
Appendix C: Supplementary Tables and Figures	28

ABSTRACT

Blood supply management is an important aspect of healthcare logistics, especially during emergencies where immediate access might save lives. Hospitals often face oscillations in the demand for various types of blood, but blood banks must efficiently distribute limited supplies. This paper proposes a low-cost multi-commodity flow model to improve blood delivery in healthcare facilities in Malolos City, Bulacan. Graph theory models blood banks and hospitals as nodes, whereas transportation routes are represented as edges with corresponding travel costs. Each blood type is considered as a separate commodity in order to confirm that compatibility constraints are satisfied. The model reduces total transportation costs while fulfilling hospital demand within supply and capacity constraints. The outcomes show that network optimization techniques can greatly increase the efficiency and reliability of blood transport systems.

Keywords: Multi-commodity flow model, Healthcare logistics, Network optimization, Blood supply management

THE PROBLEM AND ITS BACKGROUND

Health is invaluable, and many individuals rely on the help of others during medical emergencies. One of the most essential resources in healthcare is blood. Blood transfusions are necessary in many situations including trauma from accidents, surgical procedures, childbirth complications, and treatments such as chemotherapy.

However, maintaining an adequate and well-distributed blood supply remains a persistent challenge. Hospitals frequently experience shortages due to sudden spikes in demand, while blood banks must carefully manage limited inventories of different blood types. Efficient logistics therefore becomes crucial to ensure that blood reaches the facilities that need it most.

To address this challenge, this study explores the use of graph theory and network optimization in modeling blood distribution. By representing blood banks, hospitals, and transportation routes as components of a network, mathematical optimization techniques can be used to determine the most efficient distribution strategy.

Background of the Study

Blood supply management represents a critical component of healthcare infrastructure that directly impacts patient outcomes and survival rates. The complexity of blood distribution systems stems from multiple interconnected factors like, the perishable nature of blood products, the diversity of blood types with specific compatibility requirements, fluctuating demand patterns across different medical facilities, and the geographic distribution of supply and demand points.

In the Philippines, blood banking and distribution systems face particular challenges due to geographic constraints, varying levels of healthcare infrastructure development, and limited resources for logistics optimization. These challenges are often worsen by fragmented communication between blood centers, leading to difficulties in

managing and retrieving real-time data across different facilities (Eder, n.d.). Local government units and healthcare administrators often rely on manual coordination and experience-based decision-making for blood allocation, which may not always result in optimal resource utilization.

Malolos City, as the capital of Bulacan province, serves as a regional healthcare hub with multiple hospitals, clinics, and medical facilities serving both local residents and patients from surrounding municipalities. The city's blood distribution network includes major blood banks such as the Philippine Red Cross (PRC) and private blood banking facilities, along with various hospitals ranging from large provincial medical centers to smaller specialized clinics. Efforts to modernize these networks in the Philippines have included the development of web-based and SMS-integrated systems to manage blood stock inquiries and improve the efficiency of data retrieval between allied hospitals (Eder, n.d.).

Current distribution practices in the area primarily rely on direct communication between blood banks and hospitals, with allocation decisions based on immediate availability and historical demand patterns. However, this approach may not adequately account for system-wide optimization opportunities, transportation efficiency, or strategic reserve management during emergency situations. Furthermore, external factors such as the COVID-19 pandemic have highlighted significant vulnerabilities in voluntary blood donation rates and donor recruitment strategies within the Philippine context (Saleh et al., 2024).

Statement of the Problem

The objective of this study is to develop a model that optimizes the distribution of blood to hospitals across Malolos to ensure **efficient**, **equitable**, and **demand-responsive** delivery. Specifically, this study aims to:

- **Maximize total satisfied blood demand** across hospitals, subject to blood supply availability and transportation capacity constraints.

- **Minimize transportation costs** while meeting hospital demand requirements
- Evaluate Availability of blood types, **ensure that high demand blood types are prioritized** to minimize shortages during normal operating conditions

Scope and Limitation

The scope of this study encompasses the following parameters:

1. **Geographic Coverage:** Limited to healthcare facilities within Malolos City, Bulacan, including blood banks, hospitals, and clinics operating at the time of data collection.
2. **Network Components:** All identified blood banks serve as supply nodes, while hospitals and clinics serve as demand nodes in the graph model.
3. **Transportation Routes:** Primary and secondary roads commonly used for medical logistics, excluding pedestrian-only paths and non-road transport modes.
4. **Blood Types:** All eight major blood types (O+, A+, B+, AB+, O-, A-, B-, AB-) are considered as separate commodities in the multi-commodity flow model.

The study acknowledges the following limitations:

1. **Dynamic Factors:** Temporal variations such as hourly traffic fluctuations, emergency surge dynamics, and seasonal demand patterns are not incorporated in this phase of the study.
2. **External Variables:** Weather conditions, special events, and external emergencies that might affect transportation routes or demand patterns are not modeled.
3. **Blood Product Specifics:** The model focuses on whole blood units and does not differentiate between blood components (plasma, platelets, red blood cells) or special blood products.

Significance of the Study

This research contributes to multiple stakeholders and broader applications:

1. **Healthcare Administrators:** Provides a systematic framework for optimizing blood distribution decisions, potentially reducing costs and improving service reliability.
2. **Emergency Management:** Offers insights into network resilience and bottleneck identification for emergency preparedness planning.
3. **Academic Research:** Demonstrates practical application of graph theory and network optimization techniques to real-world healthcare logistics problems.
4. **Technology Integration:** Establishes a foundation for developing automated decision support systems for blood bank operations and distribution planning.

GRAPH-THEORETIC FOUNDATIONS OF THE MODEL

This section introduces the mathematical concepts necessary to understand the optimization model used in the study.

Graph Theory Framework

The distribution of blood supplies among the hospitals and the blood centers is represented into a graph theory problem to analyze and reduce blood shortages. The blood distribution system is modeled as a **directed graph** $G = (V, E)$, where each **node** represents a hospital or blood centers, and each **directed edge** represents an allowable route between facilities.

The graph is **directed** to represent the constraints in healthcare logistics, such as predefined delivery routes, policies, and non-reciprocal transportation agreements, rather than physical road directionality.

The **Nodes** in the graph are defined according to their function within the blood distribution network to represent the **Supply-Demand Structure**

- **Supply Nodes** represents blood centers that provide blood units.
- **Demand Nodes** represents hospitals or facilities that require blood units
- **Transshipment Nodes** which may both receive and redistribute blood units.

Each node maintains an inventory of the eight blood types: A+, A-, B+, B-, AB+, AB-, O-, O+.

Each weight of a **directed edge** is normalized using **distance across the nodes** as primary measure to ensure consistency across the network and avoid errors between distance-based and time-based interpretations.

The graph theory problem is addressed in this study is a **network flow** and system balance problem, where the objective is to determine the efficient distribution path and

transfer quantities that **minimize blood shortages** and to reduce preventable deaths across all nodes in the network.

Multi-Commodity Flow Model

Let $G = (V, E)$ be a directed graph, where V represents the set of vertices (or nodes) and E represents the set of directed edges (or arcs).

Capacities and Commodities

Each edge $(u, v) \in E$ is assigned a non-negative capacity, denoted by $c(u, v)$, which represents the maximum total flow that can pass through the edge. This is governed by the capacity function $c : E \rightarrow \mathbf{R}^+$.

Unlike a standard maximum flow problem, a multi-commodity flow network routes multiple distinct items, or *commodities*, through the same graph. Let $K = \{1, 2, \dots, k\}$ denote the set of commodities. For each commodity $i \in K$, the following parameters are defined:

- A source node $s_i \in V$, where the commodity originates.
- A sink node $t_i \in V$, where the commodity is destined.
- A demand $d_i > 0$, representing the required amount of flow for commodity i from s_i to t_i .

Flow Variables

The flow of a specific commodity i along an edge (u, v) is represented by the variable $f_i(u, v)$. The total flow on any given edge is the sum of the flows of all individual commodities traversing that edge.

Fundamental Constraints

A feasible multi-commodity flow must satisfy three primary mathematical conditions before any routing algorithms can be applied:

1. **Capacity Constraint:** The aggregate flow of all commodities on any edge must not exceed the capacity of that edge. For all $(u, v) \in E$:

$$\sum_{i \in K} f_i(u, v) \leq c(u, v)$$

2. **Flow Conservation Constraint:** For every intermediate node (any node that is neither the source nor the sink for a given commodity), the total incoming flow must equal the total outgoing flow. For each commodity $i \in K$ and for every vertex $u \in V \setminus \{s_i, t_i\}$:

$$\sum_{v: (v, u) \in E} f_i(v, u) - \sum_{v: (u, v) \in E} f_i(u, v) = 0$$

Furthermore, the net flow out of the source node s_i and into the sink node t_i must perfectly satisfy the demand d_i :

$$\sum_{v: (s_i, v) \in E} f_i(s_i, v) - \sum_{v: (v, s_i) \in E} f_i(v, s_i) = d_i$$

3. **Non-negativity Constraint:** The flow of any commodity along any edge cannot be negative. For all $i \in K$ and $(u, v) \in E$:

$$f_i(u, v) \geq 0$$

METHODOLOGY

The paper primarily uses the Multi-Commodity Minimum-Cost Flow Model as its center, it's a model that is generally used to optimize the transportation of multiple types of goods through a network at the lowest possible cost, while respecting constraints like capacities and demand. In this case, the Multi-Commodity Minimum-Cost Flow Model is used to to optimize the distribution of medical resources more specifically the transportation and distribution of blood to Hospitals and other facilities.

Description of the Real-Life System

The real-life system modeled after in this study consists of the healthcare infrastructure of Malolos City Bulacan, where blood banks and hospitals are represented as a complex, interconnected network. The supply side of the system is anchored by facilities like the Philippine Red Cross and local blood banks, which maintain inventories of the eight primary blood types: $O+$, $A+$, $B+$, $AB+$, $O-$, $A-$, $B-$, and $AB-$. Unlike standard flow models, this approach treats each of the eight primary blood types as a distinct commodity, ensuring that life saving units are distributed to the various hospital and blood bank. These units are distributed to various demand nodes, including major centers like Bulacan Provincial Hospital and Ace Malolos, as well as smaller clinics and maternity facilities.

Construction of the Graph Model

Dataset

Table 1*Edges in Malolos City with Supply/Demand by Blood Type*

Node ID	Facility ID	Node Type	Capacity	Supply by Type	Demand by Type
1	Red Cross	Supply	450	[158,122,104,23,9,9,4,4]	[0]
2	Blood Bank	Supply	250	[88,68,58,12,6,4,3,3]	[0]
3	Ace Malolos	Demand	70	[0]	[25,19,16,4,1,2,1,1]
4	Malolos Provincial	Demand	300	[0]	[105,81,69,15,6,6,3,3]
5	Ofelia	Demand	18	[0]	[6,5,4,1,0,1,0,1]
6	Stma Trinidad	Demand	16	[0]	[6,4,4,1,0,0,0,1]
7	Santos General	Demand	20	[0]	[7,5,5,1,0,1,0,1]
8	Sacred Heart	Demand	74	[0]	[26,20,17,4,1,2,1,1]
9	Malolos Maternity	Demand	17	[0]	[6,5,4,1,0,0,1,0]
10	Mary Immaculate	Demand	40	[0]	[14,11,9,2,1,1,1,1]
11	Malolos San Vicente	Demand	12	[0]	[4,3,3,1,0,0,1,0]
12	Romel Cruz	Demand	32	[0]	[11,9,7,2,1,1,0,1]
13	Malolos EENT	Demand	20	[0]	[7,5,1,0,1,0,1,0]
14	Graman Medical	Demand	45	[0]	[16,12,10,2,1,1,1,2]

Note: The array represents the blood type $[O+, A+, B+, AB+, O-, A-, B-, AB-]$

Edges

Table 2

Edges in Malolos City with direction/distance/travel time and cost by Nodes

Node A	Node B	Direction	Distance(km)	Travel Time(min)
1	2	$\leftrightarrow$	0.6	1.44
1	4	$\rightarrow$	0.5	1.2
2	4	$\rightarrow$	0.3	0.72
1	3	$\rightarrow$	0.7	1.68
3	7	$\leftrightarrow$	2	4.8
7	10	$\leftrightarrow$	0.4	0.96
4	5	$\leftrightarrow$	1.3	3.12
5	6	$\leftrightarrow$	0.4	0.96
3	8	$\leftrightarrow$	1.2	2.88
4	8	$\leftrightarrow$	1.1	2.64
8	9	$\leftrightarrow$	0.5	1.2
7	8	$\leftrightarrow$	1.8	4.32
7	11	$\leftrightarrow$	1.2	2.88
8	11	$\leftrightarrow$	1.3	3.12
9	13	$\leftrightarrow$	0.08	0.192
13	14	$\leftrightarrow$	1.4	3.36
13	11	$\leftrightarrow$	1.1	2.64
10	11	$\leftrightarrow$	1.4	3.36
11	14	$\leftrightarrow$	2.4	5.76
10	12	$\leftrightarrow$	4.4	10.56
12	14	$\leftrightarrow$	7.8	18.72

Modelling Assumptions and Constraints

let $G = (V, E)$ be the directed graph representing blood distribution in Malolos City where:

- V = set the nodes
- E = set of directed edges

Commodities(blood types)

$$k \in K = O+, A+ B+ AB+, O-, A-, B-, AB-$$

Supply and demand

For each node $i \in V$ and blood type $k \in K$,

$$b_{ij}^k = \begin{cases} > 0 \text{ if node } i \text{ is a blood bank (supply)} \\ < 0 \text{ if node } i \text{ is a hospital facility (demand)} \\ 0 \text{ if node } i \text{ is a transshipment facility} \end{cases}$$

Capacities

Defined as, edge cost c_{ij} is the maximum units that can be transported from node i to node j . optional type specific capacity u_{ij}^k if certain blood types are limited and Node storage capacity U_i^k for maximum inventory per blood type

Cost

The edge cost c_{ij} represents transportation effort that is based **travel time**. Normalized so that minimizing the total cost reflects the most efficient plan.

Decision Variables

$$f_{ij}^k \geq 0 \text{ for all blood type } k \text{ from } i \text{ to node } j$$

Flow Conservation Constraints

For each node i and blood type k

$$\sum_{j:(i,j) \in E} f_{ij}^k - \sum_{h:(h,i) \in E} f_{hi}^k = b_k^k$$

Edge Capacity Constraints

$$\sum_{k \in K} f_{ij}^k \leq u_{ij} \forall (i, j) \in E$$

Objective Function (Min-Cost with Priority)

Minimize total transportation cost while optionally penalizing unmet demand for a blood types

$$\sum_{(i,j) \in E} c_{ij} \sum_{k \in K} f_{ij}^k + \sum_{j \in K} \sum_{k \in K} p^k s_j^k$$

where:

- $s_j^k \geq 0$ is unmet demand (slack) for blood type k at node j
- p^k is a penalty weight for unmet demand, higher for critical types such as O-

Applied Graph-Theoretic Algorithms

The Network Simplex Method is particularly well-suited for blood supply chain optimization because it treats the logistics network as a specialized mathematical structure rather than a generic set of linear equations. By representing blood centers, distribution hubs, and hospitals as nodes connected by supply lines, the algorithm leverages the inherent properties of flow to solve complex distribution problems efficiently.

Efficiency in Multi-Objective Optimization

- **Maximize Total Satisfied Blood Demand:** The algorithm handles capacity constraints u_{ij} on every edge individually. Subject to blood supply availability and transportation limits, it ensures that flow is directed to maximize sink satisfaction.
- **Minimize Transportation Costs:** By minimizing the objective function $Z = \sum_{(i,j) \in E} c_{ij}x_{ij}$, the method identifies the most cost-effective routes while simultaneously meeting hospital demand requirements.
- **Prioritization of Blood Types:** The structure allows for the evaluation of blood type availability. High-demand or rare blood types can be prioritized through weighted cost functions to minimize shortages during normal operating conditions.

Structural Advantages

Unlike the standard Simplex algorithm, the Network Simplex maintains a **Spanning Tree** basis. This provides several advantages:

1. **Direct Mapping:** Blood centers act as source nodes, hubs as transshipment nodes, and hospitals as sink nodes.
2. **Efficiency:** It avoids heavy matrix inversions by using simple additions and subtractions across the graph edges, leading to significantly faster computation times for large-scale logistics.

RESULTS AND DISCUSSION

The researchers are successful in Creating a Multi Commodity, Minimum Cost Flow model on optimizing Blood distribution from blood banks to hospitals in Malolos City Bulacan.

The model Optimizes the routes based on minimum time needed to travel while still keeping up with the demands to maximize efficiency.

This implies that through efficiency shortages on supplies in the medical facilities that need them that are needed the most are lessened. The possibility of sudden supply drop, or demand spike also shows that shortages will be spread around since not all blood types are demanded equally, Type O+ for example is very high in demand but the limiting factor for this is more of a shortage in supply rather than a flaw in the model itself.

Events such as flood, or some roads closing may happen and can affect the distribution, but the model accounts for this that in such event there will still be an optimal route given the change on the availability of certain edges.

These results show that the use of Graph Theory can help on making or improving Models such as this Multi Commodity, Minimum Cost Flow model for the distribution of blood. Which increases the efficiency and reliability of distribution systems.

Example Calculation

To illustrate the model in practice, two example are provided. Allocations of blood units from supply nodes to demand nodes. These calculations are based on the Multi-Commodity Minimum-Cost Flow solution, showing how different blood types are distributed along selected edges while respecting supply, demand, and edge capacity constraints.

Decision variable:

f_{ij}^k = flow of blood type k from supply node i to demand node j Edge cost: c_{ij} = travel time between nodes i and j Transportation cost for each flow: $Cost_{ij}^k = f_{ij}^k c_{ij}$ Objective function: Minimize total cost while satisfying demand and capacity Minimize TotalCost

$$= \sum_{(i,j) \in E} \sum_{k \in K} (f_{ij}^k \times c_{ij}) + \sum_{j \in V} \sum_{k \in K} p_k s_j^k$$

where

s_j^k = unmet demand (slack) for blood type k at node j and

p_k = penalty weight for critical blood types

Example Calculation 1

Meeting Demand for ACE Malolos and Sacred Heart The data from the Table 1 in **Dataset** section determines how blood is distributed from the two supply nodes:

- **Red Cross** Supply: 450 units
- **Blood Bank** Supply: 250 units

to two hospitals:

- **Ace Malolos**
- **Sacred Heart**

Each hospital requires **8 blood types**.

Hospital Demand

Ace Malolos demand:

Blood Type	Demand
O+	25
A+	19
B+	16
AB+	4
O-	1
A-	2
B-	1
AB-	1

Sacred Heart demand:

Blood Type	Demand
O+	26
A+	10
B+	17
AB+	4
O-	1
A-	2
B-	1
AB-	1

Supply Availability

Red Cross supply:

Blood Type	Units
O+	158
A+	122
B+	104
AB+	23
O-	9
A-	9
B-	4
AB-	4

Blood Bank supply:

Blood Type	Units
O+	88
A+	68
B+	58
AB+	12
O-	6
A-	4
B-	3
AB-	3

Choose Routes Based on Minimum Travel Time

Route	Travel Time (min)
Red Cross → ACE Malolos	1.68
Blood Bank → ACE Malolos	2.4
Red Cross → Sacred Heart	2.64
Blood Bank → Sacred Heart	2.64

Decision: Ship from closest/fastest route first

Assign Flows

Ace Malolos(all from Red Cross, faster)

Blood Type	Flow
O+	25
A+	19
B+	16
AB+	4
O-	1
A-	2
B-	1
AB-	1

Remaining Red Cross Supply

Blood Type	Remaining
O+	133
A+	103
B+	88
AB+	19
O-	8
A-	7
B-	3
AB-	3

Sacred Heart(split between Red Cross and Blood Bank to minimize travel time)

Blood Bank → Sacred Heart (faster by 2.64 min)

Blood Type	Flow
O+	26
A+	10
B+	17
AB+	4
O-	1
A-	2
B-	1
AB-	1

Remaining Blood Bank Supply

Blood Type	Remaining
O+	62
A+	48
B+	41
AB+	8
O-	5
A-	2
B-	2
AB-	2

Transportation Cost according to previous solution:

1. **Ace Malolos** where all blood came from **Red Cross** (travel time = 1.68 min)

Blood Type	Units Shipped	Travel Time	Cost
O+	25	1.68	42
A+	19	1.68	31.92
B+	16	1.68	26.88
AB+	4	1.68	6.72
O-	1	1.68	1.68
A-	2	1.68	3.36
B-	1	1.68	1.68
AB-	1	1.68	1.68

Total cost for ACE Malolos:

$42 + 31.92 + 26.88 + 6.72 + 1.68 + 3.36 + 1.68 + 1.68 = \mathbf{11.592}$ by following the process with **Sacred Heart**:

$$68.64 + 52.8 + 44.88 + 10.56 + 2.64 + 5.28 + 2.64 + 2.64 = 190.08$$

Grand total transportation cost (ACE + Sacred Heart): $115.92 + 190.88 = \mathbf{306}$

The total transportation cost for delivering blood to Ace Malolos and Sacred Heart is 306 minutes. This value represents the optimal allocation for this scenario using the minimum-cost flow model.

This equation shows how the model optimizes the distribution of multiple commodities such as blood types having eight different types. The decision variable allows the model how to quantify the exact volume of blood needed to be transported from supply node i to demand nodes j . The first part of the objective function will calculate the Total Cost(time) by multiplying the volume of each blood type shipped by the travel time of traversing to a specific edge. The second part of the equation which deals with the penalty aspects serve as a method to minimize the new routing times affected by changes in the available edges.

This Calculates the fastest routes to transport the needed blood supply to the local hospitals that need it. The cost(time) is minimized which is why speed is being focused on to make sure that blood reaches the patients that need it as quick as possible.

and the penalty variable ensures that risks are minimized in situations where supply is too low, or demand is too high by prioritizing fulfilling orders for critical blood types.

Discussion

The results of the multi-commodity minimum-cost flow model provide awareness into how blood distribution can be optimized within Malolos City, Bulacan. By modeling blood types as separate commodities, the system ensures that allocation decisions account for compatibility and type-specific demand.

The optimized solution demonstrates how transportation routes are selected based on minimum travel time while satisfying hospital demand as efficiently as possible. Edges with high utilization indicate critical transportation corridors in the network, while facilities with unmet demand highlight potential supply shortages or capacity bottlenecks.

Sensitivity analysis further strengthens the evaluation of the model. When selected edges are removed to simulate road closures, the model reallocates flows through alternative routes, sometimes increasing total transportation cost. Unexpected scenarios, such as reduced availability from major blood banks, show how shortages spread across hospitals and which blood types are most vulnerable. Demand surge simulations for high-demand blood types like O+ reveal whether supply or transportation capacity becomes the limiting factor.

In general, the model demonstrates that graph theory and network optimization provide a systematic and data-driven framework to improve the efficiency of blood distribution. Although the current study uses a static time snapshot, the approach can be extended in future research to incorporate dynamic demand patterns, emergency scenarios, and real-time traffic conditions to further enhance system resilience.

SUMMARY AND CONCLUSION

Summary of Findings

This study developed a Multi-Commodity Minimum-Cost Flow model to optimize blood distribution across Malolos City, Bulacan using a directed, capacity-constrained network framework. By representing blood banks, hospitals, and transportation routes as a structured graph, the study demonstrated how graph theory and network flow optimization can be applied to real-world healthcare logistics.

The model ensures that blood types are treated as separate commodities, allowing compatibility constraints and type specific demand to be incorporated into the optimization process. The system finds effective allocation algorithms that prioritize essential blood types and lessen shortages among medical institutions by minimizing transportation costs and penalizing unmet demand.

Conclusions

These results show that the use of Graph Theory can help on making or improving models such as this Multi Commodity, Minimum Cost Flow model for the distribution of blood. This significantly increases the efficiency and reliability of distribution systems. Road closures, supply shocks, and demand spikes are examples of sensitivity evaluations that highlight the distribution network's resilience and point out possible bottlenecks under pressure. Overall, the findings confirm that a structured network optimization approach can significantly improve efficiency, transparency, and responsiveness in blood distribution systems. Future research may extend this framework by incorporating dynamic time based modeling, real time traffic data and emergency response planning to further enhance resilience and operational effectiveness.

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APPENDIX A

Vertex and Edge List

Table 3 and table 4 provide the comprehensive data for the healthcare facilities (vertices) and transportation routes (edges) modeled in Malolos City.

APPENDIX B

SageMath/Python Codes

The following Python code is used to process user input for solving the Multi-Commodity Flow Model and returning the optimal route:

```
1 blood_types = ['O+', 'A+', 'B+', 'AB+', 'O-', 'A-', 'B-', 'AB-']
2 supply = {
3     'Red Cross': [158, 122, 104, 23, 9, 9, 4, 4],
4     'Blood Bank': [88, 68, 58, 12, 6, 4, 3, 3],
5 }
6 demand = {
7     'ACE Malolos': [25, 19, 16, 4, 1, 2, 1, 1],
8     'Sacred Heart': [26, 10, 17, 4, 1, 2, 1, 1],
9 }
10 travel_times = {
11     ('Red Cross', 'ACE Malolos'): 1.68,
12     ('Blood Bank', 'ACE Malolos'): 2.40,
13     ('Red Cross', 'Sacred Heart'): 2.64,
14     ('Blood Bank', 'Sacred Heart'): 2.64,
15 }
16
17 def solve_blood_distribution():
18     total_cost = 0
19     results = []
20     ace_cost = 0
21     dest = 'ACE Malolos'
22     source = 'Red Cross'
23     t_time = travel_times[(source, dest)]
24
25     for idx, type_name in enumerate(blood_types):
26         units = demand[dest][idx]
27         supply[source][idx] -= units
28         cost = units * t_time
29         ace_cost += cost
30         results.append((dest, source, type_name, units, t_time,
31 cost))
32
33     sh_cost = 0
34     dest = 'Sacred Heart'
35     source = 'Blood Bank'
36     t_time = travel_times[(source, dest)]
37
38     for idx, type_name in enumerate(blood_types):
39         units = demand[dest][idx]
40         supply[source][idx] -= units
41         cost = units * t_time
42         sh_cost += cost
43         results.append((dest, source, type_name, units, t_time,
44 cost))
45
46     print(f"{'='*60}")
47     print(f"Total Cost for ACE Malolos: {ace_cost:.2f} minutes")
```

```

46     print(f"Total Cost for Sacred Heart: {sh_cost:.2f} minutes")
47
48     total_cost = ace_cost + sh_cost
49     print(f"'='*60")
50     print(f"GRAND TOTAL TRANSPORTATION COST: {total_cost:.0f}
minutes")
51     print(f"'='*60\n")
52
53     print(f"{'Destination':<15} | {'Source':<12} | {'Type':<4} | {'
Units':<5} | {'Cost'}")
54     print("-" * 60)
55     for res in results:
56         print(f"{res[0]:<15} | {res[1]:<12} | {res[2]:<4} | {res
[3]:<5} | {res[5]:.2f}")
57
58     if __name__ == "__main__":
59         solve_blood_distribution()
60

```

This function receives a 2D list representing distances between cities and evaluates all possible routes to determine the most efficient round-trip path.

1. The “Multi-Commodity” Approach

In standard flow problems, a single type of item is usually considered. However, this research treats each of the eight primary blood types (O+, A+, B+, AB+, and their negative versions) as separate commodities.

Why it matters: This ensures that the system does not simply send any available blood type to a hospital. Instead, each blood type is tracked individually to prevent mismatches, such as assigning A+ blood to a patient requiring O- blood.

Data Structure: Each hospital and blood bank maintains an array representing the exact quantity of each of the eight blood types.

2. Managing the Supply and Demand Network

The function operates on a network consisting of 14 locations in Malolos City.

Supply Nodes: These represent the origin points, specifically the Philippine Red Cross and the local Blood Bank.

Demand Nodes: These consist of 12 hospitals and clinics, including Bulacan Provincial Hospital and ACE Malolos, which require continuous blood supply.

Direct Routes: The model only allows flow along directed edges representing valid transportation routes between facilities.

3. Mathematical Rules (Constraints)

The function follows strict constraints to ensure realistic modeling:

Flow Conservation: Blood cannot be lost during transport. For intermediate nodes, the total incoming flow must equal the total outgoing flow.

Capacity Limits: Each route has a maximum allowable capacity. The total flow across all blood types must not exceed this limit.

4. The Goal: Speed and Priority

The objective of the model is to minimize total travel time. In addition, a penalty system is included to reflect emergency conditions.

Penalty for Shortages: If demand is not fully satisfied, a penalty cost is applied. Higher penalties are assigned to critical blood types such as O-.

Prioritization: Since O- is a universal donor type and often scarce, the model prioritizes fulfilling its demand first to reduce risk to patients.

5. Handling Real-World Disruptions

The model is designed to adapt to dynamic conditions in the transportation network.

Road Closures: If a route becomes unavailable due to events such as floods or accidents, the corresponding edge is removed and the model recalculates an alternative

route, even if it increases travel time.

Supply Shocks: If a blood bank experiences a shortage in a specific blood type, the system redistributes the remaining supply across hospitals in a balanced manner.

This structured approach demonstrates that transitioning from manual allocation to an automated optimization system significantly improves both the efficiency and reliability of blood distribution in Malolos City.

APPENDIX C

Supplementary Tables and Figures

Table 3

Full Vertex List (Supply and Demand Nodes)

Node ID	Facility Name	Node Type	Capacity
1	Red Cross	Supply	450
2	Blood Bank	Supply	250
3	Ace Malolos	Demand	70
4	Malolos Provincial	Demand	300
5	Ofelia	Demand	18
6	Stma Trinidad	Demand	16
7	Santos General	Demand	20
8	Sacred Heart	Demand	74
9	Malolos Maternity	Demand	17
10	Mary Immaculate	Demand	40
11	Malolos San Vicente	Demand	12
12	Romel Cruz	Demand	32
13	Malolos EENT	Demand	20
14	Graman Medical	Demand	45

Table 4

Full Edge List with Distance and Travel Time

Node A	Node B	Direction	Distance (km)	Time (min)
1	2	→	0.6	1.44
1	4	↑	0.5	1.20
2	4	↑	0.3	0.72
1	3	↑	0.7	1.68
3	7	↔	2.0	4.80
7	10	↔	0.4	0.96
4	5	↔	1.3	3.12
5	6	↔	0.4	0.96
3	8	↔	1.2	2.88
4	8	↔	1.1	2.64
8	9	↔	0.5	1.20
7	8	↔	1.8	4.32
7	11	↔	1.2	2.88
8	11	↔	1.3	3.12
9	13	↔	0.08	0.192

Continued on next page.

Table 4 (*continued*)

Node A	Node B	Direction	Distance (km)	Time (min)
13	14	$\leftrightarrow$	1.4	3.36
13	11	$\leftrightarrow$	1.1	2.64
10	11	$\leftrightarrow$	1.4	3.36
11	14	$\leftrightarrow$	2.4	5.76
10	12	$\leftrightarrow$	4.4	10.56
12	14	$\leftrightarrow$	7.8	18.72